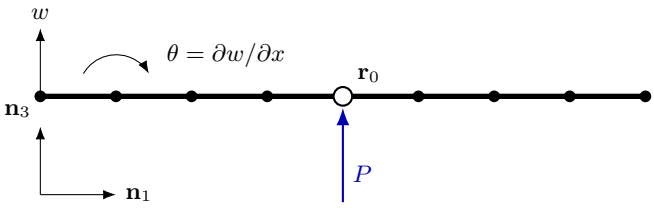
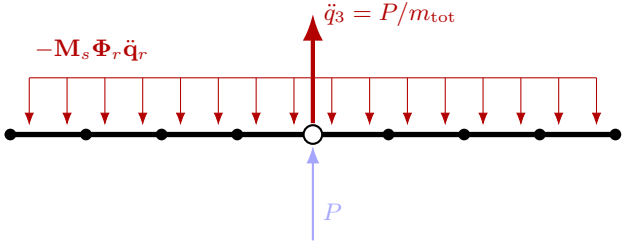
(a) Free–free beam and applied load f



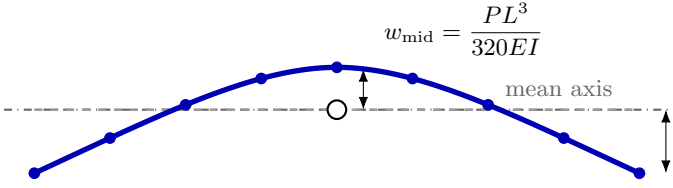
No DOFs are blanked, so $\mathbf{K}_s$ is singular. The six rigid-body modes $\mathbf{\Phi}_r$ are translations along and rotations about $\mathbf{n}_k$ through $\mathbf{r}_0$.

(b) Rigid-body acceleration and inertial load



$\mathbf{M}_{rr}\ddot{\mathbf{q}}_r = \mathbf{\Phi}_r^\top \mathbf{f}$ with $\mathbf{M}_{rr} = \mathbf{\Phi}_r^\top \mathbf{M}_s \mathbf{\Phi}_r$. The inertial load, here $-P/L$ per unit length, balances the applied load.

(c) Self-equilibrated load and elastic deformation $\mathbf{u}_e$



The bordered system solves $\mathbf{K}_s\mathbf{u}_e = \mathbf{f} - \mathbf{M}_s\mathbf{\Phi}_r\ddot{\mathbf{q}}_r$ with the mean-axis constraint $\mathbf{\Phi}_r^\top \mathbf{M}_s \mathbf{u}_e = \mathbf{0}$, giving the hogging deformation.